

Lab Worksheet R02 — Strings, Input, Conditionals & For Loops

Use: "Recitation-style Lab B" segment (40 min) of a 3-hour session. **Ground rule:** The TA never gives away answers — the TA only asks tracing questions.

Warm-up (about 5 min)

Original Practice Problem 1: *take your name as input and output the length of your name minus 5*. Fill in the blanks to complete it.

```
name = ____("Enter name: ")      # (1) built-in function that reads user input
print(____(name) - ____ )        # (2) function returning string length (3) the number
                                # to subtract
```

Run it with your own name. What number is printed, and why? _____

Core exercises (about 30 min)

Exercise 1. Remove the *n*th character from a string (original Problem 2)

(a) **Worked example** — read this original solution, then run it.

```
test_string = "We want to remove the nth character from this string"
n = 8
new_string = test_string[:n-1] + test_string[n:]
print(test_string)
print(new_string)
```

(b) **One-line change** — change only `n = 8` to `n = 3`. Predict which character disappears before running: _____

(c) **Fill in the blank** — complete the slicing so the *n*th character is removed.

```
new_string = test_string[:____] + test_string[____:]
```

[TA tracing questions]

1. "With `n = 8`, what exactly is in `test_string[:7]` at this point — how many characters?"
2. "Why `n-1` on the left slice but `n` on the right? What happens if you use `n` on both?"

Exercise 2. Length greater than 10 or less than 5? (original Problem 3)

(a) **Worked example** — read and run the original solution.

```
my_string = "This is my string" # example string - modify to test
if len(my_string) > 10 or len(my_string) < 5:
    print(True)
else:
    print(False)
```

(b) **One-line change** — change `my_string` to `"Hi"`. Predict the output before running: _____

(c) **Fill in the blank** — complete the condition.

```
if len(my_string) ____ 10 ____ len(my_string) < 5:
    print(True)
```

[TA tracing questions]

1. "For `my_string = 'This is my string'`, what does `len(my_string)` evaluate to right now?"
2. "Which side of the `or` is True for `'Hi'`? Does Python even need to check the other side?"

Exercise 3. Count the letter e with a for loop (original Problem 4)

(a) **Worked example** — read and run the original solution.

```
my_string = "How many times is the letter e in this string?"
number_of_es = 0 # this will be our counter
for letter in my_string:
    if letter == "e":
        number_of_es += 1
print(number_of_es)
```

(b) **One-line change** — change the comparison to count `"t"` instead of `"e"`. Predict the count before running: _____

(c) **Fill in the blank** — complete the loop.

```
number_of_es = 0
for ____ in my_string:
    if letter == ____:
        number_of_es ____ 1
```

[TA tracing questions]

1. "After the loop has processed the word 'times', what value is in `number_of_es`?"
2. "What does `letter` hold on the very first pass of the loop? And on the last?"

Self-check (about 5 min)

- ☐ I can slice a string with `[ :k]` and `[k:]` and predict exactly which characters each piece holds.
- ☐ I can predict the value of an `or` condition without running the code.
- ☐ I can trace a counter variable through a `for` loop, pass by pass.

Source

- MIT OpenCourseWare, **6.100L Introduction to Computer Science and Programming Using Python** (Fall 2022), **Recitation 2** (`R02_problems.py` / `R02_solutions.py`).
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